

IRAKLIS PSAROUDAKIS, Ph.D.

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Birth year: 1987
Citizenship: Greek



I am a technology enthusiast with a firm theoretical background and practical skills in software and web development. Interested in leading solutions for challenging topics in full stack system engineering, with a backend and analytics focus.

Work experience

06/2022 – now	Principal Software Engineer – Elastic, Greece (remote) Focusing on the distributed storage, network, and indexing layer of Elasticsearch. Core member of the stateless design & implementation for Serverless Elasticsearch. See our paper (DOI:10.1145/3772052.3772245), and my related YouTube (XQmbTpXfcvo) presentation. Selected public contributions: Demote cache regions for closed shard #151986, Unpromotables skip replication and peer recovery #93210, New TransportBroadcastUnpromotableAction #93600, Introduce max headroom for disk watermark stages #88639, blog post to optimize refresh costs.
01/2017 – 05/2022	Principal Member of Technical Staff (since Aug. 2018) – Oracle Labs, Zürich, Switzerland Senior Member of Technical Staff (until Jul. 2018) - In the Parallel Graph AnalytiX (PGX) team. Distributed programming & OS/runtime interaction. - Liaison for the Oracle Financial Services Crime & Compliance Studio product team, leading a team for the integration of PGX, Oracle Labs Data Studio, and innovative machine learning solutions.
08/2012 – 12/2012	Software developer (Internship) – SAP S.E., Walldorf, Germany. C++ developer for the thread and task scheduling framework of the SAP HANA database.
11/2010 – 08/2011	Computer programmer and analyst (Private; not a regular job) – Greek army, Greece. Completed my national military obligations as a private in the body of information technology.
01/2009 – 06/2010	Software and web developer (Under student fellowship; not a regular job) – Institute for the Management of Information Systems (IMIS), Athens, Greece.

Education

09/2011 – 12/2016	École Polytechnique Fédérale de Lausanne (EPFL) , Lausanne, Switzerland. Ph.D. Computer & Communication Sciences, Data-Intensive Applications (DIAS) lab. In co-op with the SAP HANA database team (Walldorf, Germany) during 01/2013-09/2016. Thesis: “Scaling Up Concurrent Analytical Workloads on Multi-Core Servers”, supervised by Prof. Anastasia Ailamaki. Research areas: - Sharing data and work across concurrent queries, to further improve performance by up to 2x. - Adaptive NUMA-aware data and task placement, to speed main-memory column-stores by up to 4x.
09/2005 – 07/2010	National Technical University of Athens (NTUA) , School of Electrical & Computer Engineering, Athens, Greece. 5-Year Diploma (MSc equivalent). Major: computer science (information systems). - GPA: 8.70/10.00. Ranking: inside 10% of the class. GPA of courses in major: 9.16/10.00. - Core modules: algorithms, web & software engineering, databases, AI, microprocessors, electronics. - Thesis “Application of Object Relational Mapping on the environment for the management of webpages of courses of NTUA”, supervised by Prof. Timos Sellis, while interning at IMIS.
06/2005	Private Senior School Pagkrito Ekpaideutirio , Heraklion, Greece. General degree 19.6/20.00.

Programming & IT skills

Excellent	- Java for backend/services, web, and desktop development. Gradle. - C++. gdb. Performance analysis with perf, Intel VTune Amplifier and PCM. - DevOps & CI/CD: Jenkins, Docker, Kubernetes. - Code & team management: BitBucket, git, Atlassian Confluence & Jira. - SQL. DBMS: Microsoft SQL Server, MySQL, PostgreSQL, Oracle, SAP HANA. - Web development: HTML, CSS, JavaScript. - Linux, bash, vim, IntelliJ IDEA, Eclipse, Visual Studio, Elasticsearch.
Good	- Cloud (AWS, GCP, Azure, OCI). Big Data (Hadoop, Hive, Spark). - Machine Learning concepts and tools. Applications & integration. Python. - LAMP (Linux, Apache, MySQL, PHP) for web development. AJAX, jQuery. - C#, .NET Framework, ASP.NET.

Selected publications and conferences

ACM SoCC, 2025	I. Psaroudakis , Pooya Salehi, Jason Bryan, Francisco Fernández Castaño, Brendan Cully, et al. “Serverless Elasticsearch: the Architecture Transformation from Stateful to Stateless” [↗] .
Devoxx Greece, 2025	I. Psaroudakis . “Elasticsearch Serverless: the Transition from Stateful to Stateless” [↗] . Public presentation about the transition of Elasticsearch’s stateful architecture to a stateless one for the serverless world.
Devoxx Greece, 2023	I. Psaroudakis . “Finding needles in ever increasing haystacks - Scalability with Elasticsearch” [↗] . Public presentation around what makes Elasticsearch distributed, scalable, efficient and resilient.
USENIX ATC, 2021	V. Trigonakis, JP Lozi, T. Faltín, N. Roth, I. Psaroudakis , A. Delamare, V. Haprian, C. Iorgulescu, P. Koupy, J. Lee, et al. “aDFS: An Almost Depth-First-Search Distributed Graph-Querying System” [↗] .
OPODIS, 2020	S. Firmli, V. Trigonakis, JP Lozi, I. Psaroudakis , A. Weld, D. Chiadmi, S. Hong, H. Chafi. “CSR++: A Fast, Scalable, Update-Friendly Graph Data Structure” [↗] .
EuroSys, 2018	I. Psaroudakis , S. Kaestle, M. Grimmer, D. Goodman, J.-P. Lozi, T. Harris. “Analytics with smart arrays: adaptive and efficient language-independent data” [↗] . (YouTube [↗])
Book, 2017	A. Ailamaki, E. Liarou, P. Tözün, D. Porobic, I. Psaroudakis . “Databases on Modern Hardware: How to Stop Underutilization and Love Multicores” [↗] . Book published by Morgan & Claypool.
VLDB, 2017	I. Psaroudakis , T. Scheuer, N. May, A. Sellami, A. Ailamaki. “Adaptive NUMA-aware data placement and task scheduling for analytical workloads in main-memory column-stores” [↗] .
TPCTC, 2014	I. Psaroudakis , F. Wolf, N. May, T. Neumann, A. Böhm, A. Ailamaki, K.-U. Sattler. “Scaling up Mixed Workloads: a Battle of Data Freshness, Flexibility, and Scheduling” [↗] .
DaMoN, 2014	I. Psaroudakis , T. Kissinger, D. Porobic, T. Ilsche, E. Liarou, P. Tözün, A. Ailamaki, W. Lehner. “Dynamic Fine-Grained Scheduling for Energy-Efficient Main-Memory Queries” [↗] .
ADMS, 2013	I. Psaroudakis , T. Scheuer, N. May and A. Ailamaki. “Task Scheduling for Highly Concurrent Analytical and Transactional Main-Memory Workloads” [↗] .
VLDB, 2013	I. Psaroudakis , M. Athanassoulis, and A. Ailamaki. “Sharing Data and Work Across Concurrent Analytical Queries” [↗] .
PC member	SIGMOD 2021 [↗] & 2024 [↗] (industrial) & 2019 [↗] (research), IEEE TPDS 2019, Voxxed Days Crete 2025 [↗] .

Selected patents

Assignee: Oracle	2026. M. Cepek, I. Psaroudakis , et al. “Graph machine learning for case similarity” [↗] . #12561215. 2025. I. Psaroudakis , et al. Address matching from single string to address matching score [↗] . #12282486. 2026. M. Cepek, I. Psaroudakis , N. Benz. “Exploiting Graph Structure to Improve Results of Entity Resolution” [↗] . Application #20230004977. 2021. A. Mumcuyan, I. Psaroudakis , M. Cepek, R. Patra. “Name matching engine boosted by machine learning” [↗] . Application #16989306. 2020. I. Psaroudakis , et al. “Language interoperable runtime adaptable data collections” [↗] . #10803087.
Assignee: SAP SE	2022. T. Scheuer, I. Psaroudakis , et al. “Adaptive table placement in NUMA architectures” [↗] . #11275721. 2020. A. Ailamaki, T. Scheuer, I. Psaroudakis , N. May. “Task scheduling for highly concurrent analytical and transaction workloads” [↗] . #10545789.

Awards and distinctions

2019	1 st team award at UK’s Fincancial Conduct Authority’s 2019 AML & Financial Crime TechSprint [↗]
2015	Ranked 2 nd place in the ACM SIGMOD programming contest [↗]
2011	Fellowship award -- awarded by the EPFL IC Doctoral School to outstanding first year PhD students.
2010	Onassis Foundation Scholarship Recipient to attend the Onassis Science Lecture Series 2010 [↗]
2007	Imagine Cup (global student competition by Microsoft) – IT Challenge – Ranked 10 th

Memberships

2023 – now	Organization member of devstaff.gr [↗] and founding member of Software Makers [↗] non-profit local orgs.
2023 – now	Volunteer for The Smile of the Child [↗] , and Positive Voice [↗] .
2024 – 2025	Organization team member for the socrates-crete.org [↗] 2024-2025 local community.
2014 – 2020	ACM (Association for Computing Machinery) member
2010 – now	Member of the Technical Chamber of Greece (TEE)
2007 – 2011	Microsoft Student Partner in Greece. Active participation with presentations, videos and training.

Personal skills

Languages	English – Fluent (Proficiency by Cambridge). French – Very good (DELFI 2 nd Degré – A5/6; C1 equiv.).	German – Conversational (Goethe B1). Greek – Native speaker.
Competences	Planning : Expertise in prioritization, budgeting, incident handling, and CAPA processes. Social : Frequent cooperation with teammates for the successful completion of projects. Problem-Solving : Worked on multiple projects which required extensive analysis and research. Ability to multitask : Worked in more than one project at the same time with success.	